

Torque Quanta & the Torsion Field — Sketch Equations (v0.1)

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Type: Speculative theoretical note

1. Idea in one line

Assume there exists a torsion field whose quanta (“torque particles”) mediate changes in spin and rotational strain, analogous to how photons mediate electromagnetic interactions.

We model this with a spin-coupled vector field .

2. Torsion / torque field definition

Let $\mathbf{S}$ be a vector field associated with rotational strain in spacetime or in an effective medium.

Define a field-strength tensor (analogous to EM):

$$F^{\{\tau\}}_{\mu\nu} = \partial_\mu \tau_\nu - \partial_\nu \tau_\mu$$

This carries the “twist” or rotational disturbance.

3. Spin current as a source

Let be an effective spin current:

$$J^{\mu}_{\text{spin}} = \partial_\nu S^{\mu\nu}$$

where is a spin density tensor (antisymmetric in).

Heuristic interpretation:

regions where spin orientation changes $\rightarrow$ nonzero .

4. Prototype Lagrangian for a torque boson

Postulate a massive vector boson (the “torque quantum”) with mass and coupling to spin current:

$$\begin{aligned} \mathcal{L}_{\tau} = & -\frac{1}{4} F^{\{\tau\}}_{\mu\nu} F^{\{\tau\}}_{\mu\nu} \\ & + \frac{1}{2} m_{\tau}^2 \tau_{\mu} \tau^{\mu} \\ & + g_{\tau} \tau_{\mu} J^{\mu}_{\text{spin}} \end{aligned}$$

First term: kinetic term for the torsion/torque field

Second: mass term (massive mediator)

Third: coupling to spin current (source of torque quanta)

5. Field equation (Euler–Lagrange)

Varying w.r.t. gives:

$$\partial_{\nu} F^{\{\tau\}}_{\nu\mu} + m_{\tau}^2 \tau^{\mu} = -g_{\tau} J^{\mu}_{\text{spin}}$$

In words: spin gradients source the torque field.

Where spin flips / rotates, torque quanta can be emitted or absorbed.

6. Effective torque density (non-relativistic limit)

In a non-relativistic, coarse-grained picture, define an effective torque density:

$$\vec{\mathcal{N}}_{\text{eff}} \approx$$

$$\gamma_{\tau} \nabla \times \vec{\tau}$$

with an effective coupling constant.

Analogy:

→ magnetic field

→ rotational “stress” or torque density

7. Energy stored in the torque field

Energy density of the torsion/torque field:

$$u_{\tau} := \frac{1}{2} \left(\vec{E}_{\tau}^2 + \vec{B}_{\tau}^2 \right) \\ \text{;+; } \frac{1}{2} m_{\tau}^2 \tau, \tau_{\mu} \tau^{\mu}$$

where we define, in analogy to EM:

$$\vec{E}_{\tau} \sim F^{\tau}_{0i}, \\ \quad \quad \quad \vec{B}_{\tau} \sim \frac{1}{2} \epsilon_{ijk} F^{\tau}_{jk}$$

Interpretation: torque quanta carry real, localizable energy associated with rotational strain.

8. Heuristic “selection rule” for torque emission

As a toy condition, a quantum transition between states $\vec{S}_i$ and $\vec{S}_f$ can emit a torque quantum if:

$$\Delta \vec{S} \neq 0$$

$$\Leftrightarrow$$

$$\exists \tau \text{ -quantum with }$$

$$\vec{J}_\tau \approx \Delta \vec{S}$$

i.e. changes in intrinsic or orbital angular momentum can be partially mediated by torque quanta instead of purely by photons or phonons in certain environments (e.g. dense spin matter, exotic phases).

9. Where might such particles show up?

This is speculative, but natural candidate regimes include:

Neutron stars / magnetars: extreme spin density and rotational shear

Topological quantum matter: skyrmions, spin textures, chiral phases

Early universe: high torsion scenarios in Einstein–Cartan-like cosmologies

Engineered metamaterials: artificial lattices with tunable spin–orbit coupling

10. Status

Consistent with field-theoretic structures (vector field + spin current coupling)

Not part of the Standard Model

Not experimentally observed

Best treated as a conceptual probe:

“What if rotational strain had its own mediator particle?”

This note is Version 0.1 — a starting point for discussion, not a final theory.